

LRF Anomaly Types

Match-Space Visualization

Shadow

$$M(r_j) \subseteq M(r_i), \alpha_i \neq \alpha_j$$

r_i : DENY

r_j : ALLOW

r_j unreachable: r_i
intercepts all $M(r_j)$
first — wrong action
applied

Fix: Triad removes r_j

Redundancy

$$M(r_j) \subseteq M(r_i), \alpha_i = \alpha_j$$

r_i : ALLOW

r_j : ALLOW

r_j unreachable;
same action as r_i
→ wastes scan
time and space

Fix: removed (Prop. 1)

Correlation

$$M(r_i) \cap M(r_j) \neq \emptyset, \text{ neither } \subseteq$$

r_i : ALLOW

r_j : DENY

overlap
(r_i wins)

order-dependent:
intersection takes r_i 's
action — not removed,
but decomposed

Fix: 4D-decompose